

Research-to-Project Intelligence Platform

STARTUP DOCUMENT · IDEA BRIEF · FULL WORKFLOW

8 Languages supported	300M+ Papers indexed	0 Direct competitors
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Prepared for: Founder's personal reference

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01

ScholarLing is an AI-powered research intelligence platform that solves three problems no existing tool solves together: finding real academic papers, understanding them in plain language, and turning them into buildable project ideas — all delivered in the user's native language.

The one-liner:

"Type a topic. Get real research papers, plain-English summaries, and concrete project ideas you can actually build — in your language."

The Problem

Students, researchers, and curious people face a painful 3-part problem when working with academic research:

01

Can't find the right papers

Google Scholar requires exact keywords. If you don't already know the terminology, you get nothing useful.

02

Can't understand what they found

Papers are written for other researchers. Dense jargon, complex methodology. Most people give up after the abstract.

03

Can't bridge research to action

Even if you understand a paper, you're stuck staring at it thinking "so what do I do with this?" Nobody tells you.

The Solution

ScholarLing collapses the entire research workflow into one intelligent pipeline. Instead of jumping between Scholar, ChatGPT, translation tools, and your own creativity, everything happens in one place:

01

Find

Real papers from PubMed, arXiv, Semantic Scholar, OpenAlex — 300M+ combined.

02

Understand

AI reads every paper and generates a plain-language abstract summary + full overview.

03 Build

AI extracts 3 concrete, buildable project ideas from each paper with tech stack + timeline.

04 In your language

Everything delivered in English, Chinese, Hindi, Japanese, Korean, Portuguese, Dutch, or French.

02

A student types a topic. ScholarLing's agent does everything else. Here is the exact pipeline from input to output:

STEP**1****User Types a Topic**

User enters any topic in any of the 8 supported languages. Natural language — no need for academic keywords. Example: "AI that helps people with hearing loss" or "solar energy storage innovations".

STEP**2****Agent Searches 4 Databases Simultaneously**

The backend agent fires parallel queries to Semantic Scholar, PubMed, arXiv, and OpenAlex. It fetches the top 10 most relevant papers by relevance score, recency, and citation count. All results are real papers with real DOI links.

STEP**3****AI Reads Every Paper**

For each paper, the AI reads the abstract (always available) and the full text where open-access content exists. It extracts: the core research question, methodology, key findings, study type, sample size, and known limitations.

STEP**4****Generates Plain-Language Summary**

The abstract is rewritten in simple language a non-researcher can understand. No jargon. No assumptions. The user immediately knows what the paper found and why it matters. Full paper overview available on expand.

STEP**5****Generates 3 Buildable Project Ideas**

This is ScholarLing's unique feature. For each paper, the AI generates 3 project ideas a student could realistically build. Each idea includes: project title, what it does, suggested tech stack, estimated build time, difficulty level, and what makes it competition-worthy.

STEP**6****Delivers Everything in User's Language**

The entire output — summaries, project ideas, citations — is delivered in the user's chosen language. A student in Japan gets everything in Japanese. A researcher in Brazil gets Portuguese. No translation step needed.

03

The most common question you will face: "Can't I just use ChatGPT or Google Scholar for this?" Here is the precise, honest answer:

Head-to-Head Comparison

Feature	Google Scholar	Consensus	ChatGPT / Claude	ScholarLing
Finds real papers	Yes	Yes	No — hallucinates	Yes — 300M+
Summarises papers	No	Yes	If you paste them	Yes — automatic
Full paper overview	No	Partial	If you paste them	Yes — on expand
Non-English papers	Limited	No	No	Yes
Output in your language	No	No	Yes (if you ask)	Yes — built-in
Generates project ideas	No	No	If you ask + paste	Yes — per paper
Saves your research history	No	Partial	No	Yes
Integrated workflow	No	Partial	No	Yes — end to end

The Real Difference

ChatGPT and Claude can summarise a paper **if you find it yourself and paste it in**. They can suggest project ideas **if you ask the right question**. But they can't search real papers, they hallucinate citations, they have no memory between sessions, and there's no structured workflow.

ScholarLing is not a chatbot. It is a **purpose-built pipeline** — search, summarise, ideate, and deliver — all automated, all in one place, all in your language. That is the product.

04

**Engineering & Science Students**

Building projects for hackathons, tech fests, and final year submissions. Need innovative ideas backed by real research but don't know where to start.

**Postgraduate Researchers**

Need to survey a new research domain quickly. Literature reviews that take weeks can be done in hours.

**Non-English Speaking Academics**

Researchers in China, India, Japan, Brazil, Korea, France, Netherlands, Portugal who are underserved by English-only tools like Consensus and Elicit.

**Journalists & Policy Makers**

Need to understand what current research says on a topic without reading 30 academic papers.

**Curious Learners**

Anyone who wants to understand what science actually says about a topic in plain language without a university login.

05

This section is your personal roadmap. Follow this phase by phase. Do not skip ahead. Each phase builds on the last.

PHASE 1 · Week 1–2

Connect the Research APIs

Goal: Get real papers appearing on screen. No AI yet. Just search working.

Tasks:

- 01 Create a free account on Semantic Scholar API (semanticscholar.org/product/api)
- 02 Create a free account on OpenAlex API (openalex.org)
- 03 Register for PubMed E-utilities API key (ncbi.nlm.nih.gov/account)
- 04 arXiv API requires no key — it's open
- 05 Build a simple Python FastAPI backend with one endpoint: POST /search
- 06 The endpoint takes a topic string, fires queries to all 4 APIs in parallel
- 07 Merge results, deduplicate by DOI, rank by relevance score
- 08 Return top 10 papers with: title, authors, year, abstract, DOI link, source
- 09 Test it with 10 different topics manually

Tech stack:

Python · FastAPI · httpx (async HTTP) · Semantic Scholar API · OpenAlex API · PubMed API · arXiv API

You're done when: You type any topic and get 10 real papers returned as JSON.

PHASE 2 · Week 3–4

Build the AI Summarisation Layer

Goal: Each paper gets a plain-language summary automatically.

Tasks:

- 01 Sign up for Claude API at console.anthropic.com (use claude-sonnet-4-6 model)

- 02 Write a summarisation prompt that takes abstract + available full text
- 03 Prompt should return: plain summary (3-4 sentences), key findings (3 bullet points), methodology type, study size, limitations
- 04 Add a second prompt for full paper overview (deeper, 8-10 sentences)
- 05 Set up Upstash Redis (free tier at upstash.com) for caching
- 06 Cache every summary by paper DOI — never summarise the same paper twice
- 07 Add the summarisation call to your /search endpoint
- 08 Each paper result now includes AI summary alongside metadata
- 09 Test quality across 20 different papers in different domains

Tech stack:

Claude API · Prompt engineering · Upstash Redis · Python

You're done when: Every paper in search results has a plain-English summary attached.

PHASE 3 · Week 5

Add the Project Idea Generator

Goal: Each paper now suggests 3 buildable student projects.

Tasks:

- 01 Write a new AI prompt: 'Given this research paper, generate 3 project ideas a student team could build in 4-8 weeks'
- 02 Each project idea must include: title, 2-sentence description, tech stack (3-4 tools), difficulty (beginner/intermediate/advanced), estimated time, and what makes it competition-worthy
- 03 Add a POST /generate-ideas endpoint that takes a paper DOI and returns ideas
- 04 Ideas should be triggered on demand (when user clicks 'Generate Project Ideas' on a paper card) — not on every search (saves API costs)
- 05 Cache generated ideas by DOI in Redis
- 06 Test with 30 papers across different domains — check that ideas are realistic and not generic

Tech stack:

Claude API · Prompt engineering · Redis caching

You're done when: Clicking any paper generates 3 specific, realistic project ideas.

PHASE 4 · Week 6

Add Multilingual Output

Goal: Everything works in all 8 languages.

Tasks:

- 01 Add a language parameter to all AI prompts: 'Respond entirely in {language}'
- 02 Add language selector UI (flag dropdown): English, Chinese, Hindi, Japanese, Korean, Portuguese, Dutch, French
- 03 Store user's language preference in their profile
- 04 All summaries, project ideas, and UI labels adapt to selected language
- 05 Test each language with 5 papers minimum — check readability with native speakers if possible
- 06 Cache summaries per language per DOI (Redis key: {doi}:{language})
- 07 Start with English + Chinese + Hindi first — highest quality and market size

Tech stack:

Claude API multilingual · Redis · React language selector

You're done when: Full search-to-ideas workflow works in all 8 languages.

PHASE 5 · Week 7

Build the Frontend

Goal: A clean, fast web app users can actually use.

Tasks:

- 01 Set up React.js project (use Vite for fast setup)
- 02 Install Tailwind CSS for styling
- 03 Build: landing page, search bar, results page, paper card component

- 04 Paper card shows: title, authors, year, source badge, plain summary, relevance score
- 05 Expandable section: full overview, key findings, limitations
- 06 Separate expandable: 3 project ideas with tech stack tags and difficulty badges
- 07 Language selector in top navigation bar
- 08 Mobile responsive — many users will be on phones
- 09 Deploy frontend to Vercel (free, connect GitHub repo, auto-deploys)

Tech stack:

React.js · Vite · Tailwind CSS · Vercel

You're done when: App looks professional, works on mobile, all features accessible.

PHASE 6 · Week 8

Add User Accounts + Search History

Goal: Users can save, bookmark, and return to past research.

Tasks:

- 01 Set up Supabase project (free at supabase.com)
- 02 Enable Google OAuth login (one toggle in Supabase dashboard)
- 03 Create database tables: users, searches, bookmarks
- 04 Save every search to user's history automatically
- 05 Add bookmark button on each paper card
- 06 Build 'My Library' page — past searches + bookmarked papers
- 07 Add language preference to user profile (saved across sessions)
- 08 Set up usage tracking per user for free tier enforcement
- 09 Deploy backend to Railway (free tier — connect GitHub, auto-deploys)

Tech stack:

Supabase · Google OAuth · PostgreSQL · Railway

You're done when: Users can log in, save searches, bookmark papers, set language preference.

PHASE 7 · Week 9–10**Payments + Beta Launch**

Goal: First paying users. Real feedback. Real revenue.

Tasks:

- 01 Set up Stripe account, create subscription products (\$12/mo Researcher, \$49/mo Institution)
- 02 Add Stripe checkout flow — users hit limit, paywall appears, one-click upgrade
- 03 Enforce free tier: 5 searches/day, abstract only, 3 languages
- 04 Paid tier: unlimited searches, full overviews, project ideas, all 8 languages
- 05 Write a Product Hunt launch post — lead with the multilingual angle
- 06 Post in Reddit communities: r/MachineLearning, r/india, r/studyabroad, r/brasil, r/korea
- 07 DM 20 professors at Indian + Brazilian universities directly on LinkedIn
- 08 Offer 3 months free to first 50 users who share honest feedback
- 09 Set up simple analytics (Plausible or Umami — privacy-friendly, free)

Tech stack:

Stripe · Product Hunt · Reddit · LinkedIn outreach

You're done when: First paying user. Feedback loop established.

	FREE	RESEARCHER \$12/month	INSTITUTION \$49/month
Searches per day	5	Unlimited	Unlimited
Abstract summary	Yes	Yes	Yes
Full paper overview	No	Yes	Yes
Project idea generator	No	Yes	Yes
Languages	3	All 8	All 8
Search history	No	Yes	Yes
Bookmarks	No	Yes	Yes
Citation export	Basic	APA/MLA/Vancouver	All formats
Team members	1	1	Up to 10
API access	No	No	Yes

Revenue Milestones

Month 1–2	Beta launch, free users only	Feedback + testimonials
Month 3	50 paid users	\$600 MRR
Month 6	200 paid users	\$2,400 MRR
Month 9	500 paid + 5 institutions	\$6,245 MRR
Month 12	1,000 paid + 15 institutions	\$12,735 MRR

07

01 Target Student Communities Directly

Join Reddit communities: r/india, r/japanlife, r/korea, r/brasil, r/MachineLearning, r/academia. Post in university WhatsApp/Telegram groups. Offer 3 months free to first 100 researchers in exchange for honest feedback. Students talk — one department spreads to the whole university.

02 Multilingual SEO — Your Biggest Moat

Write blog posts in Hindi, Chinese, Korean, and Portuguese about 'how to find research papers' and 'research paper summary tools'. Consensus and Elicit are doing zero non-English SEO. This is low-competition, high-traffic territory that compounds over time.

03 Product Hunt + Hacker News Launch

Launch on Product Hunt on a Tuesday or Wednesday morning (US time). Post a Show HN on Hacker News the same day. The multilingual + project-idea angle is genuinely novel — tech community will upvote. Target: Top 5 Product of the Day. This gets you 500–2000 signups in 48 hours.

04 University Partnerships

Email department heads at IITs, IISc, Brazilian federal universities, and Korean KAIST directly. Offer 6 months free institution plan in exchange for a testimonial and researcher survey access. One university deal = 50–200 potential paying users and major social proof.

05 Research Community Content

Create short video demos on YouTube and Instagram showing the full workflow — topic typed → papers found → project idea generated — in 60 seconds. Do this in multiple languages. Content in Hindi and Korean has almost zero competition in the research tools space.

08

Risk: Consensus launches multilingual before you ship

Mitigation: Ship in 10 weeks. First-mover advantage is real. They are a large company — they move slowly. You move fast.

Risk: AI API costs spike as users grow

Mitigation: Cache every summary aggressively by DOI + language. Same paper summarised once — free forever after. Costs stay low even at scale.

Risk: Summary quality is poor in some languages

Mitigation: Launch English, Chinese, and French first — highest AI quality. Add Hindi, Korean, Japanese in Phase 2 after testing with native speakers.

Risk: Can't access full text of paywalled papers

Mitigation: Start with abstracts only — still 10x better than competitors. Add CORE and BASE for open-access full texts progressively. Never link to pirated content.

Risk: Low conversion from free to paid

Mitigation: Make the free tier genuinely useful but clearly limited. The paywall appears exactly when users are most engaged — after finding a great paper. That is the right moment to convert.

Your Unfair Advantage in One Sentence:

You are the only platform in the world that finds real research papers, explains them in plain language, generates buildable project ideas from them, and delivers all of it in 8 languages — in a single, seamless workflow.